

1

(a) Acceleration is defined as the rate of change of velocity with respect to time.

B1 [1]

Comments: "per unit time" is NOT accepted

(b) (i) $v = u + (-g \sin \theta) t$

M1

$$= 9.0 + (-9.81 \times \sin 26^\circ) (0.70)$$

M1

$$= 5.99$$

$$= 6.0 \text{ m s}^{-1}$$

A0 [2]

(ii) Gain in GPE = Lose in KE

$$mgh = \frac{1}{2} mu^2 - \frac{1}{2} mv^2 \quad (h \text{ is the height of ramp})$$

$$gh = \frac{1}{2} u^2 - \frac{1}{2} v^2$$

$$(9.81) h = \frac{1}{2} (9.0)^2 - \frac{1}{2} (6.0)^2$$

M1

$$h = 2.3 \text{ m}$$

A1 [2]

Comments: Quite a number of candidates treat this part as projectile motion without realizing that weight is NOT the ONLY force that acts on the body. (b)(i) is a "show" question and full working needs to be shown, such as formula and full substitutions of values.

(c) (i) $\tan 5.0^\circ = v_y / v_x$

$$= v_y / (6.0 \cos 26^\circ)$$

M1

$$v_y = 0.47 \text{ m s}^{-1}$$

A1 [2]

(ii) $v_y^2 = u_y^2 + 2a_y s_y$

$$0.47^2 = (6.0 \sin 26^\circ)^2 + 2(-9.81)s_y$$

M1

$$s_y = 0.34 \text{ m}$$

A1 [2]

Comments: Error-carry-forward saved the day for a number of candidates.

(iii) sphere + ceiling (part of Earth) as system

ceiling gains upward momentum of equal magnitude as the downward momentum change of sphere (due to weight & force by ceiling)

B1

ceiling gains rightward momentum of equal magnitude as the leftward momentum change of sphere (due to friction)

B1 [2]

Comments: Candidates need to answer in details to context. Some merely says sphere transfer momentum to ceiling without saying equal magnitude and opposite direction. To gain full credit, the direction of momentum change for each object need to be stated for the context.

[11]

